Class 3

Mathematics

Solutions

Olympiad Program

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1. Numbers

Solutions

Worksheet – 1

1. **Correct Answer: (A)**
   348 = 300 + 40 + 8 → Three Hundred Forty-Eight
2. **Correct Answer: (B)**
   4,005 = 4 thousand + 5 → Four Thousand Five
3. **Correct Answer: (A)**
   1,202 = 1 thousand + 200 + 2 → One Thousand Two Hundred Two
4. **Correct Answer: (C)**
   406 = 400 + 6 → Four Hundred Six
5. **Correct Answer: (C)**
   9,876 = 9 thousand + 800 + 70 + 6 → Nine Thousand Eight Hundred Seventy-Six
6. **Correct Answer: (A)**
   7,040 = 7 thousand + 40 → Seven Thousand Forty
7. **Correct Answer: (A)**
   8,937 → Eight Thousand Nine Hundred Thirty-Seven

Worksheet – 2

1. **Correct Answer: (A)**
   Place value of 6 in 6,231
   6 is in thousands place → 6,000
2. **Correct Answer: (D)**
   Face value is 5 in 5,328 → place value
   5 is in thousands place → 5,000
3. **Correct Answer: (B)**
   Place value of 2 in 1,240
   2 is in hundreds place → 200
4. **Correct Answer: (A)**

   Sum of place values in 2,401
   = 2000 + 400 + 0 + 1 = 2,401
5. **Correct Answer: (A)**

   Difference between place value of 4 and 6 in 3,426
   4 → 400, 6 → 6
   400 − 6 = 394
6. **Correct Answer: (C)**
   Place value is 8,000 → digit must be in thousands place
   Only 8,120 has 8 in thousands place
7. **Correct Answer: (B)**
   Subtract place value of 7 from place value of 5 in 5,704
   5 → 5,000, 7 → 700
   5,000 − 700 = 4,300

Worksheet – 3

1. **Correct Answer: (A)**

   Expanded form of 256
   256 = 200 + 50 + 6
2. **Correct Answer: (A)**

   Expanded form of 812
   812 = 800 + 10 + 2
3. **Correct Answer: (A)**

   Which number is 4,000 + 600 + 0 + 3?
   = 4,603
4. **Correct Answer: (B)**

   Expanded form of 3,174
   3,174 = 3,000 + 100 + 70 + 4
5. **Correct Answer: (A)**

   Which number has expanded form 9,000 + 500 + 60 + 4?
   = 9,564
6. **Correct Answer: (B)**

   Expanded form of 7,305
   7,305 = 7,000 + 300 + 5
7. **Correct Answer: (A)**

   Expanded form of 4,802
   4,802 = 4,000 + 800 + 0 + 2

Worksheet – 4

1. **Correct Answer: (A)**
   Which is smaller: 305 or 503?
   Compare hundreds: 3 < 5 → 305 is smaller

2. **Correct Answer: (D)**
Which number is the greatest: 453, 456, 459, 467?
Compare hundreds (same), then tens → 467 is largest
3. **Correct Answer: (A)**
Descending order (largest to smallest): 654, 987, 321, 879
987 > 879 > 654 > 321
4. **Correct Answer: (C)**
Ascending order (smallest to largest): 384, 732, 527
384 < 527 < 732
5. **Correct Answer: (A)**
Descending order: 913, 931, 139, 319
931 > 913 > 319 > 139
6. **Correct Answer: (B)**
Ascending order: 812, 729, 653, 910
653 < 729 < 812 < 910
7. **Correct Answer: (A)**
Descending order: 752, 874, 618, 485
874 > 752 > 618 > 485

Worksheet – 5

1. **Correct Answer: (A) 832**
Greatest number from 3, 8, 2 (no repetition)
Arrange digits in descending order → 8, 3, 2
Number = 832
2. **Correct Answer: (A) 333**
Smallest number by repeating digits 3, 5, 7
Use the smallest digit (3) repeatedly
Number = 333
3. **Correct Answer: (D) 999999**
Greatest number by repeating digits 2, 9, 4
Use the largest digit (9) repeatedly and more digits give bigger number
Number = 999999
4. **Correct Answer: (C) 379**
Smallest number using 3, 9, 7 (all digits)
Arrange in ascending order → 3, 7, 9
Number = 379
5. **Correct Answer: (B) 409**
Smallest number using 0, 9, 4
0 cannot be at the beginning
Start with smallest non-zero digit → 4, then 0, 9
Number = 409
6. **Correct Answer: (A) 444**
Smallest number by repeating digits 4, 8, 5
Use the smallest digit (4) repeatedly
Number = 444
7. **Correct Answer: (A) 852**
Largest number from digits 5, 8, and 2 (no repetition)
Arrange digits in descending order → 8, 5, 2
Number = 852

Worksheet – 6

1. **Correct Answer: (B) 4**
Which is an even number?
Even numbers end in 0, 2, 4, 6, 8 → only 4 fits
2. **Correct Answer: (B) 15**
Which is an odd number?
Odd numbers end in 1, 3, 5, 7, 9 → 15 is odd
3. **Correct Answer: (B) 461**
Next odd number after 459
459 is odd → next odd = 459 + 2 = 461
4. **Correct Answer: (B) 836 and 902**
Which pair are both even?
Even numbers end in 0, 2, 4, 6, 8
836 and 902 are both even
5. **Correct Answer: (B) Odd**
Sum of an even and an odd number
Even + Odd = Odd
6. **Correct Answer: (B) Odd**
18 candies, give away 3
18 – 3 = 15
15 is an odd number
7. **Correct Answer: (A) 10**
Even-numbered positions out of 20 runners
Even positions: 2, 4, 6, ..., 20
Total = 10 positions

Competitive Corner

SINGLE CORRECT ANSWER TYPE

1. **Correct Answer: (C) 5,000**
5 is in the thousands place → 5,000

2. **Correct Answer: (B) 641**
   Arrange digits in descending order: 6, 4, 1 → 641
3. **Correct Answer: (A) One**
   1,052 = One thousand fifty-two
   thousand fifty-two
4. **Correct Answer: (A) 7,356**
   7,000 + 300 + 50 + 6 = 7,356
5. **Correct Answer: B**
   Break 9748 into place values:
   - Thousands = 9000
   - Hundreds = 700
   - Tens = 40
   - Ones = 8

   So,9748 = 9000 + 700 + 40 + 8
6. **Correct Answer: C**

   The number shown on the dog is 729.

   Write 729 in words:
   - 7 hundreds = Seven hundred
   - 2 tens = Twenty
   - 9 ones = Nine

   So,729 = Seven hundred and twenty nine
7. **Correct Answer: A**

   The tens place is the second digit from the right.

   Check each number:
   - 3123 → Tens digit = 2
   - 4211 → Tens digit = 1
   - 8318 → Tens digit = 1
   - 6935 → Tens digit = 3

   Only 3123 has 2 in the tens place.
8. **Correct Answer: D**

   Arrange the numbers in ascending order (smallest to largest):

   1252 (Raj) < 5425 (Tarun) < 5672 (Rahul) < 7297 (Ravi)

   So, the correct order is:
   Raj, Tarun, Rahul, Ravi
9. **Correct Answer: C**

   (i) In 78592, digit 5 is in the hundreds place → Place value = 500
   ⇒ P = 500

   (ii) Smallest 3-digit number using 5, 0, 2 (no repetition):
   First digit cannot be 0 → smallest is 205
   ⇒ Q = 205

   (iii) Sum of place values in 3015:
   3000 + 0 + 10 + 5 = 3015
   ⇒ R = 3015
10. **Correct Answer: C**

    To form the largest 4-digit number, arrange the digits in descending order:

    7 > 5 > 1 > 0

    So, the number formed is 7510
11. **Correct Answer: C**

    Break 4759 according to place values:
    4 → thousands = 4000
    7 → hundreds = 700
    5 → tens = 50
    9 → ones = 9

    So, expanded form = 4000 + 700 + 50 + 9
12. **Correct Answer: D**

    To form the greatest 4-digit number, arrange the digits in descending order:

    7 > 3 > 2 > 1

    So, the number formed is 7321
13. **Correct Answer: B**
    A number is even if its last digit is even (0, 2, 4, 6, 8).
    After interchange, the ones place becomes the thousands digit.
    Check each option:
    A. 5562 → after interchange → 2565 → last digit = 5 (odd)
    B. 4493 → after interchange → 3494 → last digit = 4 (even)
    C. 7734 → after interchange → 4737 → last digit = 7 (odd)
    D. 7257 → after interchange → 7257 → last digit = 7 (odd)
14. **Correct Answer: B. Kamal**

    Compare the numbers:
    Ali = 1570
    Rishi = 1270
    Kamal = 2185

    2185 is the greatest number.

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

15. **Correct Answers: (A), (B)**
    Odd numbers end in 1, 3, 5, 7, 9 → 145, 503
16. **Correct Answers: (A), (C)**
    Even numbers end in 0, 2, 4, 6, 8 → 124, 680
17. **Correct Answers: (A), (C)**
    Greater than 732 → 842, 759
18. **Correct Answers: (A), (C), (D)**
    Descending order (largest to smallest):
    875 > 764 > 653
    912 > 803 > 701
    980 > 889 > 788
19. **Correct Answers: (A), (C)**
    4-digit numbers → 1,002 and 9,876

ASSERTION AND REASON TYPE

20. **Correct Answer: (C)**
    Assertion is correct (4,602 → Four thousand six hundred two).
    Reason is false (commas are not used to write number names).
21. **Correct Answer: (A)**
    5 is in thousands place → 5,000
    Reason is also correct and explains it.
22. **Correct Answer: (A)**
    Smallest 4-digit number from 3, 0, 6, 2 → 2,036
    Reason is correct (arrange in ascending order, keep 0 after first digit).

MATRIX MATCHING TYPE

23. **Correct Answer: (a)-(Q), (b)-(S), (c)-(R), (d)-(P)**
    A. 5,000 + 200 + 30 + 6 = 5,236 → (Q)
    B. Face value of 8 = 8 → (S)
    C. Largest 3-digit number = 999 → (R)
    D. 7,341 in words → Seven thousand three hundred forty-one → (P)
24. **Correct Answer: (a)-(P), (b)-(Q), (c)-(R), (d)-(S)**
    A. 7 is in thousands place → 7,000 → (P)
    B. Largest 4-digit without repeating → 9,876 → (Q)
    C. Smallest 3-digit even number → 100 → (R)
    D. 1,204 = 1,000 + 200 + 4 → (S)

COMPREHENSION TYPE

Passage 1:

25. **Correct Answer: (C) 6,00,000**
    6 is in lakhs place → 6,00,000
26. **Correct Answer: (A)**
    6,47,320 = Six lakh forty-seven thousand three hundred twenty
27. **Correct Answer: (C) 4**
    Face value means the digit itself → 4

Passage 2:

28. **Correct Answer: (C) 935**
    Odd numbers end in 1, 3, 5, 7, 9 → 935
29. **Correct Answer: (C) 800**
    8 is in hundreds place → 800
30. **Correct Answer: (A) 824**
    Even numbers end in 0, 2, 4, 6, 8 → 824

2. Basic operations

Worksheet – 1

1. **Correct Answer: (B) 557**
   Solution: 245+312=557.
2. **Correct Answer: (A) 379**
   Solution: 123+256=379.
3. **Correct Answer: (A) 565**
   Solution: 436+129=565.
4. **Correct Answer: (A) 789**
   Solution: 343+456=789.
5. **Correct Answer: (A) 886**
   Solution: 732+154=886.
6. **Correct Answer: (A) 1791**
   Solution: 1324+567=1791.
7. **Correct Answer: (A) 3997**
   Solution: 2435+1562=3997.

Worksheet-2

1. **Correct Answer: (A) 132**
   Solution: 245−123=132.
2. **Correct Answer: (C) 221**
   Solution: 432−211=221.
3. **Correct Answer: (A) 211**
   Solution: 532−321=211.
4. **Correct Answer: (A) 456 km**
   Solution: 1245−789=456.
5. **Correct Answer: (B) 285**
   Solution: 674−389=285.
6. **Correct Answer: (B) 909**
   Solution: 2432−1523=909.
7. **Correct Answer: (A) 1755**
   Solution: 5000−3245=1755.

Worksheet-3

8. **Correct Answer: (A) 375**
   Solution: 125×3=375.
9. **Correct Answer: (C) 864**
   Solution: 432×2=864.
10. **Correct Answer: (B) 860**
    Solution: 215×4=860.
11. **Correct Answer: (B) 2070**
    Solution: 345×6=2070.
12. **Correct Answer: (A) 1404**
    Solution: 468×3=1404.
13. **Correct Answer: (B) 4536**
    Solution: 567×8=4536.
14. **Correct Answer: (A) 53935**
    Solution: 2345×23=53935.

Worksheet- 4

15. **Correct Answer: (B) 6**
    Solution: 18÷3=6.
16. **Correct Answer: (B) 7**
    Solution: 42÷6=7.
17. **Correct Answer: (B) 7**
    Solution: 28÷4=7.
18. **Correct Answer: (C) 12**
    Solution: 96÷8=12.
19. **Correct Answer: (C) 8**
    Solution: 72÷9=8.
20. **Correct Answer: (C) 12**
    Solution: 144÷12=12.
21. **Correct Answer: (C) 16**
    Solution: 256÷16=16.

Worksheet-5

1. **Correct Answer: (B) 20**
   Solution:
   Multiply first: 4×2=8.
   Then add: 12+8=20.
2. **Correct Answer: (B) 6**
   Solution:
   Solve inside the brackets: 10+2=12.
   Then divide: 12÷2=6.
3. **Correct Answer: (A) 10**
   Solution:
   Subtract first: 50−10=40.
   Then divide: 40÷4=10.
4. **Correct Answer: (B) 16**
   Solution:
   Divide first: 18÷3=6.
   Then multiply: 4×2=8.
   Finally, add: 6+8=16.
   Solution:
   Solve inside the parentheses first: 6÷2=3.
   Subtract: 15−3=12.
   Multiply: 12×3=30.
5. **Correct Answer: (C) 22**
   Solution:
   Divide first: 144÷12=12.

Then add: 12+10=22.

6. **Correct Answer: (B) 90**
   Solution:
   Solve inside the parentheses: 10÷2=5.
   Subtract: 30−5=25.
   Add: 20+25=45.
   Multiply: 45×2=90.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: B) 623**

   Solution:
   245 + 378 = 623

2. **Correct Answer: A) 735**

   Solution:
   1200 - 465 = 735

3. **Correct Answer: A) 48**

   Solution:
   6 × 8 = 48

4. **Correct Answer: B) 8**

   Solution:
   56 ÷ 7 = 8

5. **Correct Answer: B) 65**

   Solution:
   50 + 25 - 10 = 65

6. **Correct Answer: D. 7346**

   Greatest + Smallest 4-digit number (digits 3, 0, 6, 1)
   Greatest number = 6310
   Smallest number = 1036
   Sum = 6310 + 1036 = 7346

7. **Correct Answer: C. >**

   Compare and fill the box
   Left side = 1579 + 2504 − 978 = 3105
   Right side = 845 + 1840 − 1050 = 1635
   Since 3105 > 1635

8. **Correct Answer: D. 1846**

   Apples sold on both days
   First day = 983
   Second day = 983 − 120 = 863
   Total = 983 + 863 = 1846

9. **Correct Answer: C. 3838**

   Total seats in the train
   Total = 1298 + 2540 = 3838

10. **Correct Answer: C. 7438**

    Identify the 4-digit number
    Ones digit = largest 1-digit even number = **8**
    Tens digit = **8 − 5 = 3**
    Hundreds digit = ones digit of **758 − 354 = 404** = **4**
    Thousands digit = **3 + 4 = 7**

    Number = **7438**

11. **Correct Answer:D. 5**

    Find the value of △
    ○ + ○ + ○ = 150
    So, each ○ = 150 ÷ 3 = 50

    Given, ○ ÷ △ = **10**
    So, **50 ÷ △ = 10**
    △ = **50 ÷ 10 = 5**

12. **Correct Answer: C. 4182**

    Total children visited the library
    First week = 3153
    Second week = 1029

    Total = 3153 + 1029 = 4182

13. **Correct Answer: D. (p) 2202, (q) 1045**

    13. Train journey

    (q) After leaving station A
    Passengers left = 1560 − 515 = 1045

    (p) Passengers now
    At station B, 1157 boarded
    Total now = 1045 + 1157 = 2202

14. **Correct Answer: D. 24**

    Find the value of P + Q

    Given:

    - 2 × 9 = 18
    - 3 × P = 18 → P = 18 ÷ 3 = 6
    - Q × 1 = 18 → Q = 18

    So,
    P + Q = 6 + 18 = 24

15. **Correct Answer: B. 25**

    Hair bands distributed among 5 girls

    Total hair bands = **125**
    Girls = **5**

Each girl gets = **125 ÷ 5 = 25**

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

16. **Correct Answers: A) Add 15 and C) Multiply by 2.**

A) 87 + 15 = 102 (greater than 100)
B) 87 - 5 = 82 (not greater than 100)
C) 87 × 2 = 174 (greater than 100)
D) 87 ÷ 3 = 29 (not greater than 100)

17. **Correct Answers: A) Add 50 and C) Multiply by 2.**

Solution:
A) 50 + 50 = 100 (correct)
B) 50 - 50 = 0 (not correct)
C) 50 × 2 = 100 (correct)
D) 50 ÷ 2 = 25 (not correct)

18. **Correct Answer: B) Subtraction.**

A) Addition (not applicable)
B) Subtraction (90 - number given away) (correct)
C) Multiplication (not applicable)
D) Division (not applicable)

19. **Correct Answer: A) Add 30.**

Solution:
A) 20 + 30 = 50 (correct)
B) 20 × 3 = 60 (not correct)
C) 20 + 20 - 5 = 35 (not correct)
D) 20 - 10 = 10 (not correct)

20. **Correct Answer: A) Subtract 60.**

Solution:
A) 100 - 60 = 40 (correct)
B) 100 ÷ 2 = 50 (not correct)
C) 100 × 0.5 = 50 (not correct)
D) 100 + 10 = 110 (not correct)

ASSERTION AND REASON TYPE

21. **Correct Answer: Both Assertion (A) and Reason (R) are true, and Reason correctly explains Assertion.**

Solution:

Assertion (A):
267 + 468 = 735
→ 267 + 468 = 735 (Correct)
Reason (R):
Addition is the process of combining two or more numbers to get a total. (Correct definition)

22. **Correct Answer: A) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).**

Solution:

- **Assertion (A)** is false (125 - 75 = 50 is correct).
- **Reason (R)** is true.

23. **Correct Answer: D) Assertion (A) is false, and Reason (R) is true.**

Assertion (A) is false (7 × 8 = 56).

Reason (R) is true.

MATRIX MATCHING TYPE

24. **Correct Answer:**
(a) → (S), (b) → (R), (c) → (Q), (d) → (P)

Solution:

- 123+456=579
- 789+234=1023
- 346+678=1024
- 567+453=1020

25. **Correct Answer:**
(a) → (P), (b) → (Q), (c) → (S), (d) → (R)

Solution:

- 345+456−123=678
- (234×3)÷2=351
- 678−345+234=567
- (567+432)−(123×2)=753

COMPREHENSION TYPE

26. **Correct Answer: C. ₹480**
Money spent on apples
Apples = 2 kg × ₹240
= ₹480

27. **Correct Answer: B. ₹860**
Total money spent on all fruits
Apples = **₹480**
Bananas = **3 × ₹60 = ₹180**
Grapes = **₹200**
Total = 480 + 180 + 200 = ₹860

28. Correct Answer: B. ₹200

Money left after buying watermelon

Money left = **₹500**
Watermelon cost = **₹300**

Remaining = 500 − 300 = ₹200

Passage 2: School Picnic Planning

29. Correct Answer: C. ₹10,000

Total money collected

Students = **25**
Contribution per student = **₹400**

Total = 25 × 400 = ₹10,000

30. Correct Answer: C. ₹8,500

Total money spent

Bus rental = **₹5,000**
Food = **₹2,500**
Games = **₹1,000**

Total spent = 5000 + 2500 + 1000 = ₹8,500

31. Correct Answer: B. ₹1,500 and 15 students

Money left for return gifts

Total collected = **₹10,000**
Total spent = **₹8,500**

Remaining = 10,000 − 8,500 = ₹1,500

Each gift = **₹100**

Students receiving gifts = **1500 ÷ 100 = 15**

3. Fractions

Solutions

Worksheet – 1

1. **Correct Answers: (A) 3**
   The numerator shows the number of shaded parts.
   There are 3 shaded parts
2. **Correct Answers: (A) 2**
   The part being selected in a fraction is shown by the **numerator**.
   In $\frac{2}{5}$, the selected part is **2**
3. **Correct Answers: (C) 3/4**
   The circle is divided into 4 equal parts, and 3 parts are shaded.
   So the fraction is $\frac{3}{4}$
4. **Correct Answers: (A) 4/6**
   The rectangle is divided into 6 equal parts.
   2 parts are shaded, so 4 parts are not shaded.
   Fraction not shaded = $\frac{4}{6}$
5. **Correct Answers: (A) 2/8**
   The pie chart has **8 equal parts**.
   There are **2 blue parts (light + dark blue)**.
   Fraction = $\frac{2}{8}$
6. **Correct Answers: (B) 3/4**
   The square is divided into 4 equal parts.
   3 parts are shaded.
   Fraction shaded = $\frac{3}{4}$
7. **Correct Answers: (A) 7/12**
   The rectangle has 12 equal parts.
   5 parts are shaded, so 7 parts are not shaded.
   Fraction not shaded = $\frac{7}{12}$

Worksheet – 2

1. **Correct Answers: (B) 2/8**
   The circle is divided into 8 equal parts.
   2 parts are shaded.
   Fraction = $\frac{\mathbf{2}}{\mathbf{8}}$
2. **Correct Answers: (A) ¼**
   There are 4 apples in total.
   1 apple is green.
   Fraction = $\frac{\mathbf{1}}{\mathbf{4}}$
3. **Correct Answers: (A) 6/8**
   The rectangle is divided into 8 equal parts.
   6 parts are shaded.
   Fraction = $\frac{\mathbf{6}}{\mathbf{8}}$
4. **Correct Answers: (A) 3/6**
   There are 6 balls in total.
   3 balls are red.
   Fraction = $\frac{\mathbf{3}}{\mathbf{6}}$
5. **Correct Answers: (B) 2/8**
   The pie chart has 8 equal parts.
   2 parts are unshaded.
   Fraction = $\frac{\mathbf{2}}{\mathbf{8}}$
6. **Correct Answers: (C) 4/12**
   Total stars = 12, differently colored = 4.
   Fraction = $\frac{\mathbf{4}}{\mathbf{12}}$
7. **Correct Answers: (B) 2/4, 3/4, 1, and 5/4**
   The number line is divided into equal parts of $\frac{\mathbf{1}}{\mathbf{4}}$.
   Starting from 0:
   - First mark = $\frac{\mathbf{1}}{\mathbf{4}}$
   - Next marks continue as $\frac{\mathbf{2}}{\mathbf{4}}, \frac{\mathbf{3}}{\mathbf{4}}, \mathbf{1}, \frac{\mathbf{5}}{\mathbf{4}}$

   So:
   - M = $\frac{\mathbf{2}}{\mathbf{4}}$
   - N = $\frac{\mathbf{3}}{\mathbf{4}}$
   - = $\mathbf{1}$
   - P = $\frac{\mathbf{5}}{\mathbf{4}}$

Worksheet – 3

1. **Correct Answers: (A) 4/5**
   Proper fraction → numerator < denominator
   Only $\frac{\mathbf{4}}{\mathbf{5}}$fits

2. **Correct Answers: (A) Improper Fraction**
   Numerator > denominator → Improper fraction
3. **Correct Answers: (B) Improper Fraction**
   $\frac{12}{8}$ → numerator > denominator
4. **Correct Answers: (C) A whole number**
   Denominator = 1 → value is a whole number
5. **Correct Answers: (C) 1 as numerator**
   A unit fraction has numerator = 1
6. **Correct Answers: (C) Both same**
   $\frac{3}{6} = \frac{1}{2}$ → both equal
7. **Correct Answers: (C) Unlike Fractions**
   $\frac{2}{3}$ and $\frac{3}{5}$ have different denominators

Worksheet – 4

1. **Correct Answers: (D) 5/4**
   Compare by cross multiplication:
   For $\frac{1}{2}, \frac{1}{3}, \frac{2}{4}, \frac{5}{4}$
   Compare $\frac{5}{4}$ with others:
   - $\frac{5}{4}$ vs $\frac{1}{2}$:
     $5 \times 2 = 10,\ 1 \times 4 = 4 \Rightarrow 10 > 4$
   - $\frac{5}{4}$ vs $\frac{1}{3}$:
     $5 \times 3 = 15,\ 1 \times 4 = 4 \Rightarrow 15 > 4$
   - $\frac{5}{4}$ vs $\frac{2}{4}$:
     $5 \times 4 = 20,\ 2 \times 4 = 8 \Rightarrow 20 > 8$
2. **Correct Answers: (A) First student**
   Compare $\frac{5}{6}$ and $\frac{4}{5}$:
   Cross multiply:
   $$5 \times 5 = 25$$
   $$4 \times 6 = 24$$
   Since $25 > 24$,
   $$\frac{5}{6} > \frac{4}{5}$$
3. **Correct Answers: (B) Second bottle**
   Same denominator → compare numerators:
   $$3 > 2$$
4. **Correct Answers: (C) Both same**
   $\frac{4}{6} = \frac{2}{3}$ → equal
5. **Correct Answers: (B)**
   Ascending: $\frac{3}{10}, \frac{5}{10}, \frac{7}{10}$
   Same denominator (10) → compare numerators:
   $$3 < 5 < 7$$
6. **Correct Answers: (B)**
   Same denominator (3) → compare numerators:
   $$1 < 2 < 3$$
   → Ascending: $\frac{1}{3}, \frac{2}{3}, \frac{3}{3}$
7. **Correct Answers: (B)**
   Same numerator (4) → larger denominator = smaller fraction:
   $$\frac{4}{10} < \frac{4}{8} < \frac{4}{5}$$

Worksheet – 5

1. **Correct Answers: (D) 4/9**
   Same denominator → subtract:
   $$\frac{5}{9} - \frac{1}{9} = \frac{4}{9}$$
2. **Correct Answers: (A) 5**
   Add:
   $$\frac{1}{2} + \frac{9}{2} = \frac{10}{2} = 5$$
3. **Correct Answers: (B) 9/12**
   Same denominator → add:
   $$\frac{7}{12} + \frac{2}{12} = \frac{9}{12}$$
4. **Correct Answers: (A) 5/6**
   $\frac{4}{6} + \frac{1}{6} = \frac{5}{6}$
5. **Correct Answers: (B) 1/2**
   $\frac{1}{8} + \frac{3}{8} = \frac{4}{8} = \frac{1}{2}$
6. **Correct Answers: (A) 4/10**
   $\frac{7}{10} - \frac{3}{10} = \frac{4}{10}$
7. **Correct Answers: (C)**
   Incorrect one:
   $\frac{4}{6} + \frac{1}{6} = \frac{5}{6}$, not $\frac{5}{12}$

COMPETITIVE CORNER

SINGLE CORRECT ANSWER TYPE

1. **Correct Answers: (C) $\frac{6}{8}$**
   All fractions have the same denominator (8), so compare numerators.
   Only 6 is greater than 5.
2. **Correct Answers: (A)**
   $\frac{1}{3}$ means 1 out of 3 equal parts.
   (A) Correct → $\frac{1}{3}$
   (B) $\frac{1}{4}$

(C) $\frac{1}{5}$

(D) $\frac{3}{6} = \frac{1}{2}$

3. **Correct Answers: A.** $\frac{10}{16}$

Total number of small squares = 16
Number of **unshaded** squares = 10

Fraction of unshaded part = $\frac{10}{16}$

4. **Correct Answers: (B)** $\frac{5}{11}$

Total number of students = 11
Number of boys = 5

Fraction of boys = $\frac{5}{11}$

5. **Correct Answers:(A)** $\frac{4}{5}$

Total eaten = $\frac{1}{5} + \frac{2}{5} + \frac{1}{5}$
$= \frac{4}{5}$

6. **Correct Answers: C.(p)** $\frac{3}{11}$ **(q)** $\frac{2}{11}$

Total alphabets = 11

(p) Vowels = A, U, I → 3 → $\frac{3}{11}$

(q) X appears 2 times → $\frac{2}{11}$

7. **Correct Answers: C.** $\frac{7}{19}$

Total eaten = $5 + 3 + 4 = 12$
Strawberries left = 7

Total strawberries = $12 + 7 = 19$

Fraction left = $\frac{7}{19}$

8. **Correct Answers: D.** $\frac{3}{17}, \frac{5}{17}, \frac{9}{17}, \frac{13}{17}$

All fractions have the same denominator (17), so compare numerators.
Ascending order: 3 < 5 < 9 < 13

So, correct order = $\frac{3}{17}, \frac{5}{17}, \frac{9}{17}, \frac{13}{17}$

9. **Correct Answers: A. R, P, Q**

Count shaded parts in each figure:

- **P**: 4 out of 8 parts shaded → $\frac{4}{8}$
- **Q**: 5 out of 8 parts shaded → $\frac{5}{8}$
- **R**: 3 out of 8 parts shaded → $\frac{3}{8}$

Ascending order: $\frac{3}{8} < \frac{4}{8} < \frac{5}{8}$
So → **R, P, Q**

10. **Correct Answers: C**

Figure A: The triangle is divided into 4 equal smaller triangles. 2 of them are shaded. The fraction is 2/4 (which simplifies to 1/2).

Figure B: The pentagon is divided into 5 equal sections. 2 of them are shaded. The fraction is 2/5.

Figure C: The circle is divided into 3 equal sections. 2 of them are shaded. The fraction is 2/3.

Figure D: The rectangle is divided into 6 equal squares. 5 of them are shaded. The fraction is 5/6.

11. **Correct Answers: C**

A: Same denominator → 5 < 7 < 9 Correct
B: Vowels in CREATION = C R E A T I O N → E, A, I, O (4 vowels out of 8) → $\frac{4}{8}$ Correct
C: Even numbers from 1 to 15 = 2, 4, 6, 8, 10, 12, 14 → 7 numbers → $\frac{7}{15}$, not $\frac{8}{15}$
Incorrect

12. **Correct Answers: (A)**

Left = $\frac{20}{20} - \frac{9}{20} = \frac{11}{20}$

13. **Correct Answers: A.** $\frac{3}{10}$

Total birds = 10
Sparrows + pigeons = $4 + 3 = 7$
Parrots = $10 - 7 = 3$

Fraction of parrots = $\frac{3}{10}$

14. **Correct Answers: C.** $\frac{9}{15}$

Total mangoes = 15
Given away = 6
Left = $15 - 6 = 9$

Fraction left = $\frac{9}{15}$

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

15. **Answers: A, C**

Proper fractions → numerator < denominator

$\frac{3}{5}, \frac{7}{8}$

16. **Answers: A, C**

Equivalent to $\frac{1}{2}$

$$\frac{2}{4} = \frac{1}{2}, \frac{4}{8} = \frac{1}{2}$$

17. **Answers: A, B, D**

Improper → numerator ⩾ denominator

$$\frac{7}{4}, \frac{3}{2}, \frac{9}{9}$$

18. **Answers: A, B**

Same denominator → compare numerators
Correct order: $2 < 5 < 7$
So A and B are correct

19. **Answers: A, D**

Less than 1 → numerator < denominator

$$\frac{3}{4}, \frac{1}{3}$$

ASSERTION AND REASON TYPE

20. **Correct Answers: (B)**

Assertion: To compare $\frac{3}{4}$ and $\frac{2}{3}$:

Multiply crosswise:

$$3 \times 3 = 9$$
$$2 \times 4 = 8$$

Since $9 > 8$,

$$\frac{3}{4} > \frac{2}{3}$$

Conclusion: Assertion is True

Reason: True (rule is correct), but denominators are different here, so it does not explain the assertion.

21. **Correct Answers: (C)**

Assertion: $\frac{5}{6}$ → numerator < denominator → Proper fraction

Reason: Says numerator ⩾ denominator → This is wrong (this is improper fraction)

22. **Correct Answers: (C)**

Assertion: Correct definition

Reason: Mixed number is always greater than 1, not less

MATRIX MATCHING TYPE

23. **Correct Answers: (a)–P, (b)–Q, (c)–S, (d)–R**

A. numerator < denominator → Proper Fraction → P

B. diagram → Visual Representation → Q

C. numerator ⩾ denominator → Improper Fractions → S

D. same denominator → Like Fraction → R

24. **Correct Answers: (a)–P, (b)–R, (c)–Q, (d)–S**

A. $\frac{1}{2} + \frac{3}{2} = \frac{4}{2} = 2 \rightarrow$ P

B. $\frac{3}{4} - \frac{1}{4} = \frac{2}{4} = \frac{1}{2} \rightarrow$ R

C. $\frac{1}{3} + \frac{2}{3} = \frac{3}{3} = 1 \rightarrow$ Q

D. $\frac{5}{6} - \frac{3}{6} = \frac{2}{6} = \frac{1}{3} \rightarrow$ S

COMPREHENSION TYPE

Passage 1:

25. **Correct Answers: C**

$\frac{4}{9} \rightarrow$ numerator < denominator → Proper fraction

26. **Correct Answers: C**

$\frac{11}{9} \rightarrow$ numerator > denominator → Improper fraction

27. **Correct Answers: D**

$1\frac{2}{9} \rightarrow$ whole number + fraction → Mixed fraction

Passage 2:

28. **Correct Answers: A**

Total eaten = $\frac{3}{10} + \frac{2}{10} = \frac{5}{10}$

29. **Correct Answers: B**

Cake left = $\frac{10}{10} - \frac{5}{10} = \frac{5}{10}$

30. **Correct Answers: C**

Same denominator (10) → Like fractions

4. Time and money

Solutions

Worksheet – 1

1. **Correct Answer: (C) Tuesday**
   Thursday + 5 days → Friday (1), Saturday (2), Sunday (3), Monday (4), Tuesday (5)
2. **Correct Answer: (C) 240**
   4 hours = 4 × 60 = **240 minutes**
3. **Correct Answer: (C) 72**
   3 days = 3 × 24 = **72 hours**
4. **Correct Answer: (C) June 20th**
   June 5 + 15 days = **June 20**
5. **Correct Answer: (C) Both are the same**
   3 hr 45 min = (3×60)+45 = 180+45 = **225 min**
6. **Correct Answer: (B) 3 hours and 15 minutes**
   195 min ÷ 60 = 3 hr (180 min) + 15 min → **3 hr 15 min**
7. **Correct Answer: (A) 54**
   2 days = 48 hr; 48 + 6 = **54 hr**

Worksheet – 2

1. **Correct Answer: (D) 06:30**
   6:30 AM in 24-hour format = **06:30**
2. **Correct Answer: (C) 22:00**
   7:00 PM = 19:00; 19:00 + 3 hr = **22:00**
3. **Correct Answer: (A) 20:45**
   8:45 PM = 20:45
4. **Correct Answer: (A) 13:00**
   London 18:00 − 5 hr = **13:00**
5. **Correct Answer: (B) 2:05 AM**
   23:15 + 2 hr 50 min = 02:05 → **2:05 AM**
6. **Correct Answer: (B) 09:10**
   08:30 + 40 min = **09:10**
7. **Correct Answer: (A) 6:00 AM**
   22:45 + 7 hr 15 min = 06:00 → **6:00 AM**

Worksheet – 3

1. **Correct Answer: (B) 11:45 AM**
   Start = 9:15 AM
   - 2 hours → 11:15 AM
   - 30 minutes → **11:45 AM**
2. **Correct Answer: (B) 1 hour 15 minutes**
   3:00 PM to 4:00 PM = 1 hour
   4:00 PM to 4:15 PM = 15 minutes
   Total = **1 hour 15 minutes**
3. **Correct Answer: (B) 11:55 AM**
   Start = 8:40 AM
   3 hours → 11:40 AM
   15 minutes → **11:55 AM**
4. **Correct Answer: (B) 7:45 PM**
   End = 10:30 PM
   Go back 2 hours→ 8:30 PM
   Go back 45 minute→ **7:45 PM**
5. **Correct Answer: (B) 3 hours 45 minutes**
   Start = 2:45 PM
   2:45 → 3:45 = 1 hour
   3:45 → 4:45 = 1 hour
   4:45 → 5:45 = 1 hour
   5:45 → 6:30 = 45 minutes
   Total = 3 hours 45 minutes
6. **Correct Answer: (A) 12:15 PM**
   Arrival = 5:45 PM
   − 5 hours → 12:45 PM
   − 30 minutes → **12:15 PM**
7. **Correct Answer: (B) 3 hours 45 minutes**
   Start = 6:20 AM
   6:20 → 9:20 = 3 hours
   9:20 → 10:05 = 45 minutes
   Total = **3 hours 45 minutes**

Worksheet – 4

1. **Correct Answer: (C) 5,00,000 paise**
   ₹1 = 100 paise
   ₹5,000 = 5,000 × 100 = **5,00,000 paise**
2. **Correct Answer: (A) 12,575 paise**
   ₹125.75 × 100 = **12,575 paise**
3. **Correct Answer: (B) 50**
   ₹100 ÷ ₹2 = **50 coins**
4. **Correct Answer: (C) ₹20 + ₹20 + ₹5**
   Check options:
   ₹20 + ₹20 + ₹5 = **₹45**

5. **Correct Answer: (C) ₹15**
   Paid = ₹20, Change = ₹5
   Cost = 20 − 5 = **₹15**
6. **Correct Answer: (A) ₹10 + ₹2 + 50p**
   ₹12.50 = ₹12 + 50 paise
   Check options:
   ₹10 + ₹2 + 50p = **₹12.50**
7. **Correct Answer: (B) 150 paise = ₹1.50**
   100 paise = ₹1
   150 paise = 150 ÷ 100 = **₹1.50**

Worksheet – 5

1. **Correct Answer: (B) ₹37.75**
   ₹25.50 + ₱12.25 = **₹37.75**
2. **Correct Answer: (A) ₹2,250.25**
   ₹1,250.50 + ₱999.75 = **₹2,250.25**
3. **Correct Answer: (B) ₹79.25**
   ₹200 − ₱120.75 = **₹79.25**
4. **Correct Answer: (A) ₹103.00**
   ₹25.75 × 4 = **₹103.00**
5. **Correct Answer: (A) ₹2,103.00**
   ₹350.50 × 6 = **₹2,103.00**
6. **Correct Answer: (A) ₹12.50**
   ₹62.50 ÷ 5 = **₹12.50**
7. **Correct Answer: (B) ₹143.00**
   ₹2,145 ÷ 15 = **₹143.00**

Competitive Corner

SINGLE CORRECT ANSWER TYPE

1. **Correct Answer: (B) Friday**
   Thursday is followed by Friday.
2. **Correct Answer: (C) 72 hours**
   1 day = 24 hours ⟶ 3 × 24 = 72 hours
3. **Correct Answer: (B) 3:30 PM**
   15:30 in 12-hour format = 3:30 PM
4. **Correct Answer: (B) 2 hours 30 minutes**
   From 6:15 PM to 8:45 PM = 2 hours 30 minutes
5. **Correct Answer: (A) 15 minutes**
   Next hour is 11:00 ⟶ 10:45 to 11:00 = 15 minutes
6. **Correct Answer: (B) 300 minutes**
   1 hour = 60 minutes ⟶ 5 × 60 = 300 minutes
7. **Correct Answer: (B) Toy Train – ₹950**

   Add the money shown: ₹500 + ₹200 + ₹100 + ₹100 + ₹20 = ₹920

   Compare prices:
   A: ₹710 (can buy)
   B: ₹950 (more than ₹920, cannot buy)
   C: ₹650
   D: ₹520
8. **Correct Answer: (B) (B) ₹1123**
   Price of one charger = ₹199 ⟶ 2 × 199 = ₹398
   Price of one stand = ₹145 ⟶ 5 × 145 = ₹725
   Total = 398 + 725 = ₹1123
9. **Correct Answer: (B) ₹1655**
   T-shirt = ₹515
   Pants = ₹690
   Shoes = ₹450
   Total = 515 + 690 + 450 = ₹1655
10. **Correct Answer: (D) 50**
    Cost of bicycle = ₹500
    Savings per week = ₹10
    Weeks needed = 500 ÷ 10 = 50 weeks
11. **Correct Answer: (C) ₹216**
    Cost of 1 bottle = ₹36
    Cost of 6 bottles = 36 × 6 = ₹216
12. **Correct Answer: (D) ₹65**
    Compare all prices: 195, 180, 85, 65
    Smallest = ₹65
13. **Correct Answer: (C) ₹160**
    Cost of 1 chair = ₹850 ⟶ 2 chairs = 850 × 2 = ₹1700
    Cost of 1 table = ₹640
    Total cost = 1700 + 640 = ₹2340
    Money given = ₹2500
    Change = 2500 − 2340 = ₹160
14. **Correct Answer: A, B, C**
    From 9:30 AM to 2:00 PM
    = 4 hours 30 minutes
    - True (4 hr 30 min)
    - True (2:00 PM is afternoon)
    - True (9:30 AM is morning)
    - False
15. **Correct Answer: A, B, C, D**
    Check 24-hour conversions:
    - 8:30 PM = 20:30
    - 6:00 AM = 06:00
    - 3:15 PM = 15:15
    - 10:45 PM = 22:45
16. **Correct Answer: A, B, C, D**
    Find total = ₹5
    - 2 + 2 + 1 = 5
    - 1 × 5 = 5

- 5 × 1 = 5
- (2×2) + (0.50×2) = 4 + 1 = 5

17. **Correct Answer: A, C, D**
    2 hours 30 minutes = $\mathbf{2 \times 60 + 30 = 120 + 30 = 150}$minutes
    (A) 150 minutes
    (B) 90 minutes
    (C) 120 + 30 = 150
    (D) 2 × 60 + 30 = 150
18. **Correct Answer: A, B, C, D**
    1 rupee = 100 paise
    (A) 2 × 50 = 100
    (B) 4 × 25 = 100
    (C) 10 × 10 = 100
    (D) 5 × 20 = 100
19. **Correct Answer: A, C, D**
    12:15 AM is just after midnight
    - Morning (early morning)
    - Midnight is 12:00 AM, not 12:15
    - Comes after 12:00 midnight
    - 00:15 in 24-hour format

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

20. **Correct Answer: B**
    Assertion (A): A day has 24 hours True
    - Reason (R): There are 60 minutes in an hour True
    - But R does not explain why a day has 24 hours
21. **Correct answer: C**
    Assertion (A): 5 days = $\mathbf{5 \times 24 = 120}$hours True
    - Reason (R): There are 12 hours in a day False (There are 24 hours)
22. **Correct answer: A**
    Assertion (A): Change = 50 – 20 = 30 True
    - Reason (R): Change is found by subtracting price from amount paid True
    - R correctly explains A

ASSERTION AND REASON TYPE

23. **Correct answer: (a)–P, (b)–R, (c)–Q, (d)–S**
    - 7:45 (PM) in 24-hour format = 19:45 → (P)
    - ₹5,000 = 5,000 × 100 = 5,00,000 paise → (R)
    - 1 minute = 60 seconds → (Q)
    - 24 hours = 24 × 60 = 1440 minutes → (S)
24. **Correct answer: (a)–S, (b)–Q, (c)–R, (d)–P**
    - 2 hours = 2 × 60 = 120 minutes → (S)
    - 8:30 AM is in the morning → (Q)
    - 2000 ÷ 5 = 400 → (R)
    - 4 days = 4 × 24 = 96 hours → (P)

COMPREHENSION TYPE

Passage 1:

25. **Correct answer: C) ₹1900**
    Total cost = 500 + 800 + 600 = ₹1900
26. **Correct answer: C) 2 hours**
    Given: Time spent playing games = 2 hours
27. **Correct answer: B) 1 hour 30 minutes**
    Total party time = 4:00 PM to 7:30 PM = 3 hours 30 minutes
    Time spent playing = 2 hours
    Remaining time = 3 hr 30 min – 2 hr = 1 hr 30 min

Passage 2:

28. **Correct answer: B) 2 hours 30 minutes**
    Time = 6:00 PM to 8:30 PM = 2 hours 30 minutes
29. **Correct answer: C) $120**
    Total budget = $200
    Spent = $80
    Remaining = 200 – 80 = $120
30. **Correct answer: C) $20**
    Remaining money = $120
    Each prize = 120 ÷ 6 = $20

5. Measurements

Solutions

Worksheet – 1

1. **Correct Answer: (A) 100 cm**
   Solution: 1 meter = 100 centimeters.
2. **Correct Answer: (C) Millimeter**
   Solution: The order of units from smallest to largest is: millimeter → centimeter → meter → kilometer.
3. **Correct Answer: (B) 6 pieces**
   Solution: 1 meter = 100 cm, so 3 meters = 300 cm. Each piece is 50 cm.
   Number of pieces = 300/50=6
4. **Correct Answer: (B) Meter**
   Solution: The metric system uses the **meter** as the standard unit of length.
5. **Correct Answer: (A) 150 mm**
   Solution: 1 cm = 10 mm, so 15 cm = 15×10=150 mm.
6. **Correct Answer: (B) 24 meters**
   Solution: Perimeter of a rectangle = 2×(Length+Width)
   2×(8+3)=2×11=24 meters
7. **Correct Answer: (A) 1.3 meters**
   Solution: 3.5−2.2=1.3 meters.

Worksheet-2

1. **Correct Answer: (C) Square meter**
   Solution: The area is measured in square units, and in the metric system, the most common unit of area is the square meter (m^2).
2. **Correct Answer: (B) 15 m^2**
   Solution: The area of a rectangle is calculated using the formula:
   $$Area = Length \times Width$$
   So, $5\,m \times 3\,m = 15\ m^2$.
3. **Correct Answer: (B) 40 m^2**
   Solution: The area of the garden is calculated as:
   $$Area = Length \times Width$$
   $$10\,m \times 4\,m = 40\ m^2$$
4. **Correct Answer: (A) 53.8 ft^2**
   Solution: To convert square meters to square feet, multiply by 10.76:
   $$5\ m^2 \times 10.76\,ft^2/m^2 = 53.8 ft^2$$
5. **Correct Answer: (A) 48 m^2**
   Solution: The area of the pool is:
   $Area = Length \times Width$
   8 m×6 m=48 m^2
6. **Correct Answer: (B) 2800 m^2**
   Solution:
   Area of the plot: $50\,m \times 30\,m = 1500\ m^2$
   New dimensions of the plot with pathway:
   $$Length = 50 + 2 \times 1 = 52\,m$$
   $$Width = 30 + 2 \times 1 = 32\,m$$
   Area with pathway:
   $$52\,m \times 32\,m = 1664\ m^2$$
   Area of the pathway:
   $$1664\ m^2 - 1500\ m^2 = 164\,m^2.$$
7. **Correct Answer: (B) 8 m**
   Solution: To find the length of a side of a square, take the square root of the area:
   $$Side = \sqrt{64} = \sqrt{8^2} = 8\,m$$

Worksheet-3

8. **Correct Answer: (B) Gram**
   Solution: The basic unit of mass in the metric system is the gram (g). A kilogram (kg) is equal to 1000 grams.
9. **Correct Answer: (C) 5 kg**
   Solution: To find the total mass, you add the two amounts together: 3 kg + 2 kg = 5 kg.
10. **Correct Answer: (C) Both weigh the same**
    Solution: One kilogram of feathers and one kilogram of bricks both weigh the same: 1 kilogram.
11. **Correct Answer: (A) 1.5 kg**
    Solution: To convert grams to kilograms, divide by 1000: 1500 g ÷ 1000 = 1.5 kg.
12. **Correct Answer: (A) 2 kg**
    Solution: Subtract the used amount from the total: 2.5 kg - 0.5 kg = 2 kg.

13. **Correct Answer: (A) 6000 g**
Solution: First, add the weights in kilograms: 4 kg + 2 kg = 6 kg. Then, convert to grams: 6 kg × 1000 = 6000 g.
14. **Correct Answer: (B) 3500 g**
Solution: Convert 1 kg to grams: 1 kg = 1000 g. Now, add: 2500 g + 1000 g = 3500 g.

Worksheet-4

15. **Correct Answer: (A) Liter**
Solution: The basic unit of volume in the metric system is the liter (L), used to measure liquids.
16. **Correct Answer: (B) Graduated cylinder**
Solution: A graduated cylinder is specifically designed to measure the volume of liquids and can also help in measuring the volume of irregular solids using water displacement.
17. **Correct Answer: (B) 60 cm^3**
Solution: The volume of a box (rectangular prism) is calculated by multiplying length × width × height. So, 4 cm × 3 cm × 5 cm = 60 cm^3.
18. **Correct Answer: (C) 1000**
Solution: There are 1000 milliliters (mL) in 1 liter (L), as the metric system defines the conversion between these units.
19. **Correct Answer: (A) 1.5 L**
Solution: If you pour out 500 mL from 2 L, you have 2 L - 0.5 L = 1.5 L remaining.
20. **Correct Answer: (C) 150000 mL**
Solution: Since 1 liter is equal to 1000 milliliters, we multiply 150 L × 1000 mL/L = 150000 mL.
21. **Correct Answer: (C) 27 cm^3**
Solution: The volume of a cube is calculated as side × side × side. For a cube with a side length of 3 cm, the volume is 3 cm × 3 cm × 3 cm = 27 cm^3.

Worksheet-5

22. **Correct Answer: (A) 0°C**
Solution: Water freezes at **0°C** on the Celsius scale.
23. **Correct Answer: (A) 212°F**
Solution: Water boils at **212°F** on the Fahrenheit scale.
24. **Correct Answer: (B) 77°F**
Solution: The formula for Celsius to Fahrenheit is:
$F=(C\times95)+32$
Substitute C=25:
$F=(25\times95)+32=45+32=77$ °F
25. **Correct Answer: (B) Celsius**
Solution: Celsius is widely used in science because it is part of the metric system.
26. **Answer: (B) 5°C**
Solution: The formula for Fahrenheit to Celsius is:
$C=5/9\times(F-32)$
Substitute F=40:
$C=5/9\times(40-32)=5/9\times8=5$ °C
27. **Correct Answer: (A) 277 K**
Solution: The formula for Celsius to Kelvin is:
$K=C+273$
Substitute C=4:
$K=4+273=277$ K.
28. **Correct Answer: (A) 37°C**
Solution: The normal human body temperature is approximately **37°C**.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Answer: B. Centimeter**

Solution: The length of small objects like a pencil is measured in centimeters. Kilometers are for long distances, kilograms for weight, and liters for volume.

2. **Correct Answer: C. 1,000**

Solution: The metric system states that 1 kilometer equals 1,000 meters.

3. **Correct Answer: (D)**

7:30 – 1 hr 15 min = 6:15

4. **Correct Answer: (C)**
More than 50°C and less than 70°C = 60°C
5. **Correct Answer: (A)**
(5 × 2) + (2 × 3) = 10 + 6 = 16 kg
6. **Correct Answer: B. 8 L 220 mL**

Solution: 15 L 500 mL – 7 L 280 mL = 8 L 220 mL

7. **Correct Answer: D. 40 minutes**
   Solution: 7:10 p.m. – 6:30 p.m. = 40 minutes
8. **Correct Answer:** D. 2195 cm
   Solution: 439 × 5 = **2195 cm**
9. **Correct Answer:** A. 3 kg 475 g
   Solution: 7 kg 595 g – 4 kg 120 g = **3 kg 475 g**
10. **Correct Answer:** A. 4
    Solution: 28 ÷ 7 = **4 weeks**

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

11. **Correct Answer: A. Meter, B. Kilometer, D. Millimeter**

    Solution: Metric units for length include meter, kilometer, and millimeter. Gram is a unit of mass.

12. **Correct Answer: A. Liter, B. Milliliter, C. Cubic Meter**

    Solution: Liter, milliliter, and cubic meter are units for measuring volume

13. **Correct Answer: B. Water boils, C. Steam is formed**

    Solution: At 100°C, water reaches its boiling point and starts turning into steam.

14. **Correct Answer: A. Millimeter, B. Centimeter, C. Decimeter**

    Solution: Submultiples of a meter are millimeter, centimeter, and decimeter. Kilometer is a multiple of a meter.

15. **Correct Answer: A. It is measured in square units, B. It can be calculated as length × width, D. It is used to describe flat surfaces.**

    Solution: Area is measured in square units, calculated using length × width, and describes flat surfaces

ASSERTION AND REASON TYPE

16. **Correct Answer: (A) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).**

    Explanation: In the metric system, a kilometer is a multiple of a meter, and 1 kilometer = 1,000 meters.

17. **Correct Answer: (C) Assertion (A) is true, and Reason (R) is false.**

    Explanation: The area of a rectangle is calculated as length × width, but area is always measured in square units, not cubic units.

18. **Correct Answer: (C) Assertion (A) is true, and Reason (R) is false.**

    Explanation: Mass is measured using a weighing scale, but it refers to the amount of matter in an object, not the space it occupies

MATRIX MATCHING TYPE

19. **Correct Answer:**
    (a) → (S), (b) → (R), (c) → (P), (d) → (Q)
    Explanation:
    - Kilogram (kg) is a unit of mass.
    - Liter (L) is a unit used to measure volume.
    - Celsius (°C) is a unit used to measure temperature.
    - Meter (m) is a unit of length in the metric system.
20. **Correct Answer:**
    (a) → (R), (b) → (P), (c) → (Q), (d) → (S)
    Explanation:
    - 1 kilometer = 1,000 meters.
    - 1 liter = 1,000 milliliters.
    - 1 meter = 100 centimeters.
    - 1 kilogram = 1,000 grams.

COMPREHENSION TYPE

Passage-1

21. **Correct Answer: (B) 11 kilograms**

    Solution: 4 kg (watermelon) + 2 kg (bananas) + 5 kg (potatoes) = **11 kilograms**

22. **Correct Answer: (A) 2 liters**

    Solution: Total capacity of the jar = 5 liters. Milk poured = 3 liters. Remaining capacity = **5 - 3 = 2 liters**.

23. **Correct Answer: (B) 5°C**

    Solution: Temperature in sunlight = 30°C, temperature in the shade = 25°C. Difference = **30 - 25 = 5°C**.

    Passage-2

24. **Correct Answer: (A) 30 cubic meters**

    Solution: Volume = Length × Width × Depth = **5 × 3 × 2 = 30 cubic meters**.

25. **Correct Answer: (C) 30 square meters**

    Solution: The area to be covered by grass is given as **30 square meters**.

26. **Correct Answer: (A) 89.6°F**

    Solution: Fahrenheit = (°C × 9/5) + 32 = (32 × 9/5) + 32 = **89.6°F**.

6. Geometry

Solutions

Worksheet – 1

1. **Correct Answer: (B) Point**
   A location with no length or width is called a **point**. It only shows position.
2. **Correct Answer: (B) Line segment**
   A part of a line that has **two endpoints** is called a **line segment**.
3. **Correct Answer: (B) Line segment**
   The straight path between two points A and B is a **line segment**.
4. **Correct Answer: (C) Line**
   A line extended in both directions is called a **line**.
5. **Correct Answer: (B) Line segment**
   A tight rope between two poles has two endpoints, so it represents a **line segment**.
6. **Correct Answer: (A) Line Segment**
   The edge of a book is straight and has two ends, so it is a **line segment**.
7. **Correct Answer: (C) Line Segment**
   A pencil is straight and has two ends, so it represents a **line segment**.

Worksheet – 2

1. **Correct Answer: (B) 3**
   A triangle has **3 sides**.
2. **Correct Answer: (C) Square**
   A plane shape is flat; a **square** is flat.
3. **Correct Answer: (B) Solid shape**
   A cylinder is a 3D (solid) shape.
4. **Correct Answer: (B) Cone**
   A **cone** has a circular base.
5. **Correct Answer: (A) Cone**
   A shape with 1 circular base and 1 curved surface is a **cone**.
6. **Correct Answer: (A) A circle**
   A sphere looks like a **circle** from any direction.
7. **Correct Answer: (B) 6 faces, 12 edges, 8 vertices**
   A dice (cube) has 6 faces, 12 edges, and 8 vertices.

Worksheet – 3

1. **Correct Answer: (B) Sphere**
   Curved surface
   Only a sphere has a curved surface.
2. **Correct Answer: (B) Rectangle**
   Side view of cylinder
   From the side, it looks like a rectangle.
3. **Correct Answer: (A) Cube**
   Box with 6 square faces
   A cube has 6 square faces.
4. **Correct Answer: (A) Rectangle**
   Side view of triangular prism
   From the side, it looks like a rectangle.
5. **Correct Answer: (B) The curved surface**
   Rolling a cylinder
   A cylinder rolls on its curved surface.
6. **Correct Answer: (A) A rectangle**
   Top view of a rectangular prism
   Top view of a rectangular prism is a rectangle.
7. **Correct Answer: (B) Square**
   Top view of a cube
   From the top, a cube looks like a square.

Worksheet – 4

8. **Correct Answer: (C)**
   Shape with 5 faces, 5 vertices, 8 edges
   A square pyramid has 5 faces, 5 vertices, and 8 edges.
9. **Correct Answer: (B) 12**
   Edges in a cuboid
   A cuboid has 12 edges.
10. **Correct Answer: (B) 6**
    Faces of a rectangular prism
    A rectangular prism has 6 faces.
11. **Correct Answer: (A) 0**
    Edges of a sphere
    A sphere has no edges.
12. **Correct Answer: (B) 4**
    Vertices of a triangular pyramid
    It has 4 vertices.

13. **Correct Answer: (C) Rectangular Prism**
Box with 2 square faces and 4 rectangular faces
This matches a rectangular prism.
14. **Correct Answer: (B) 1**
A cone has one vertex (the pointed top).

Worksheet – 5

15. **Answer: (C) 16 meters**
Perimeter of rectangle
Perimeter = 2 × (length + width) = 2 × (5 + 3) = 16
16. **Answer: (C) 16 meters**
Perimeter of square
Perimeter = 4 × side = 4 × 4 = 16
17. **Answer: (C) 8 meters**
Third side of triangle
Perimeter = 24, two sides = 8 + 8 = 16
Third side = 24 − 16 = 8
18. **Answer: (B) 24 meters**
Boundary of square park
Perimeter = 4 × 6 = 24
19. **Answer: (D) 44 meters**
Boundary of rectangular garden
Perimeter = 2 × (15 + 7) = 2 × 22 = 44
20. **Answer: (B) 20π meters**
Circumference of circle
Circumference = 2πr = 2π × 10 = 20π
21. **Answer: (D) 52 cm**
Add all outer sides:
- Bottom = 15 cm
- Right side = 9 cm
- Top right = 9 cm
- Small vertical = 3 cm
- Top left = 5 cm
- Left side = 11 cm

Total = 15 + 9 + 9 + 3 + 5 + 11 = 52 cm

COMPETITIVE CORNER

SINGLE CORRECT ANSWER TYPE

1. **Answer: (A) A dot made by a pencil**
A point is shown by a small dot.
2. **Answer: (A) 40 meters**
Perimeter of rectangle = 2 × (length + breadth)
= 2 × (12 + 8) = 2 × 20 = 40 m
3. **Answer: (A) Circle**
Base of a cone is always a circle.
4. **Answer: (C) 28 cm**
Perimeter of triangle = sum of all sides
= 7 + 9 + 12 = 28 cm
5. **Correct Answer: C. (p)-(b); (q)-(c); (r)-(a)**
Cuboid → 6 faces, 8 vertices → (b)
- Cone → 2 faces, 1 vertex → (c)
- Sphere → 1 face, 0 vertex → (a)

6. **Correct Answer: (D) Square**
A 2D shape with 4 equal sides is a square.
7. **Correct Answer: (D) There are 4 cones.**
From the figure:
- Cylinders = 4
- Squares = 6
- Circles = 3
- Cones = 3 (not 4)

8. **Correct Answer: (D) Cylinder**
Cube → only flat faces
- Pyramid → only flat faces
- Triangular pyramid → only flat faces
- Cylinder → has a curved surface

9. **Correct Answer: (B) 0**
The given figure has no corners (vertices).
10. **Correct Answer: (A) 45 cm**
Perimeter of triangle = sum of all sides
= 15 + 15 + 15 = 45 cm
11. **Correct Answer: (A) 10**
Squares = 4
Triangles = 6
Total = 4 + 6 = 10
12. **Correct Answer: (B) 42 cm**
Add all outer sides:
= 7 + 3 + 3 + 5 + 2 + 3 + 6 = **29 cm**
Bottom :3+7+3=13
Add both 29+13=42

13. **Correct Answer: (A) 44 cm**
The board is a rectangle.
Perimeter = 2 × (length + breadth)
= 2 × (15 + 7)
= 2 × 22 = 44 cm
14. **Correct Answer: (C) 20 meters**
Perimeter of square = 4 × side
= 4 × 5 = **20 meters**

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

15. **Correct Answer: A, B, C**
Cylinder, sphere, and cone have curved surfaces, but a cube has only flat faces.
16. **Correct Answer: A, B, D**
Football field, book cover, and circular plate are flat (2D), while a dice is a solid (3D).
17. **Correct Answer: A, B, D**
Both have 6 faces, 8 vertices, and 12 edges, and no curved surface.
18. **Correct Answer: A, C**
A rectangle has opposite sides equal and 4 corners. All sides equal is for a square.
19. **Correct Answer: A, D**
Cube and cuboid have straight edges, while cone and cylinder have curved surfaces.

ASSERTION AND REASON TYPE

20. **Correct Answer: A**

Assertion (A): A line segment is part of a line that has two endpoints.
Reason (R): A line continues indefinitely in both directions without endpoints.

Both statements are true, and the reason correctly explains why a line segment has endpoints (it is a part of a line).

21. **Correct Answer: B**

Assertion (A): A cube is a solid shape with six square faces.
Reason (R): A triangle is a plane shape with three edges.

Both statements are true, but the reason does not explain the assertion.

22. **Correct Answer: C**

Assertion (A): The top view of a cube shows a square.
Reason (R): The front view of a cube shows three faces.

Assertion is true (top view of a cube is a square).
Reason is false (front view shows only one face, not three).

MATRIX MATCHING TYPE

23. **Correct Answer: (a)–P, (b)–Q, (c)–R, (d)–S**
    A. Front view of a cube → (P) Square
    B. Top view of a cylinder → (Q) Circle
    C. Side view of a rectangular prism → (R) Rectangle
    D. Front view of a triangular prism → (S) Triangle
24. **Correct Answer:(a)–S, (b)–Q, (c)–P, (d)–R**
A hexagonal prism has 8 faces → (a) → (S)
    - A cube has 12 edges → (b) → (Q)
    - A pyramid (square pyramid) has 5 faces → (c) → (P)
    - A tetrahedron has 4 vertices → (d) → (R)

COMPREHENSION TYPE

Passage 1:

25. **Correct Answer: C. 6**
The box is a cube, and a cube has 6 faces.
26. **Correct Answer: C. 12**
A cube has 12 edges.
27. **Correct Answer: C. Vertex**
Edges meet at a point called a vertex.

Passage 2:

28. **Correct Answer: C. Rectangle**
The passage clearly says the garden is rectangular.
29. **Correct Answer: B. 12 meters**
Given length = 12 meters.
30. **Correct Answer: D. 32 meters**
Perimeter = 2 × (Length + Width)
= 2 × (12 + 4) = 2 × 16 = 32 meters

7. Data Handling

Solutions

Worksheet – 1

PICTOGRAPH

1. **Correct Answer: (A) 5**
   Explanation: Each 🍌 represents 1 banana. There are 5 bananas in total.
2. **Correct Answer: (A) 3**
   Explanation: Each 🍎 symbol represents 1 apple, so 3 apples are sold.
3. **Correct Answer: (B) 12**
   Explanation: 3 apples, 5 bananas, and 2 oranges = 3+5+2=12.
4. **Correct Answer: (A) 12**
   Explanation: Each 🍇 represents 2 grapes. There are 6 symbols, so 6×2=12.
5. **Correct Answer: (A) 15**
   Explanation: Each 🚗 represents 3 cars. There are 5 symbols, so 5×3=15.
6. **Correct Answer: (D) 14**
   Explanation: 4 apples + 3 bananas + 5 oranges = 4+3+5=14.
7. **Correct Answer: (B) 12**
   Explanation: Each ✏️ represents 2 pencils. There are 6 symbols, so 6×2=12.

Worksheet-2

TALLY MARKS

1. **Correct Answer: (A) 𝍸 ||||**

   Explanation: A group of 5 𝍸 and 2 more || make 7.

2. **Correct Answer: (A) 13**

   Explanation: 𝍸 = 5, 𝍸 = 5, ||| = 3; 5+5+3=13.

3. **Correct Answer: (C) 18**

   Explanation:

   - 🧸: 𝍸 |||=8
   - 🚗: 𝍸 𝍸 = 10

   Total: 8+10=18.

4. **Correct Answer: (D) 𝍸 𝍸 𝍸 ||||**
   Explanation: 𝍸 = 5, 𝍸 = 5, |||| =4; 5+5+4=19.
5. **Correct Answer: (A) 13**
   Explanation: 𝍸 = 5, 𝍸 = 5, ||| =3; 5+5+3=13.
6. **Correct Answer: (B) 5**

   Explanation:

   - 🐶: 𝍸 || =7
   - 🐱: 𝍸 𝍸 || =12

   Difference: 12−7=5.

7. **Correct Answer: B) 5**

   Explanation: Twenty-five items would include five sets of five.

Worksheet-3.

BAR GRAPHS AND SCALING

1. **Correct Answer: (A) Apples**

   Explanation: The bar for apples 🍎 is the tallest, showing the highest value.

2. **Correct Answer: (A) 10**

   Explanation: Each unit represents 2 books, so 5×2=10.

3. **Correct Answer: (A) 50**

   Explanation: Add the values: 15+10+20+5=50.

4. **Correct Answer: (A) 20**

   Explanation: If the bar for 🍪 is twice as tall, they sold 10×2=20 cookies.

5. **Correct Answer: (B) 30**

   Explanation: 6×5=30.

6. **Correct Answer: (A) 5 units**

   Explanation: 50/10=5

7. **Correct Answer: (A) 5**

   Explanation: 25/5=5 symbols.

COMPETITIVE CORNER

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: (B) 20**

   Solution: Each 🍎 represents 5 apples. Therefore, 4×5=20 apples were sold.

2. **Correct Answer: (C) 11**
   Solution: Each group of 4 vertical marks with a diagonal line represents 5 pencils. So, 5+5+1=11.

3. **Correct Answer: C. 60**
   From the bar graph:
   - Monday = **70**
   - Tuesday = **50**
   - Wednesday = **60**

   First, find Monday + Tuesday: $70 + 50 = 120$
   Now find how many less on Wednesday: $120 - 60 = 60$

4. **Correct Answer: A. 165**
   Children on all four days:
   Monday = 70
   Tuesday = 50
   Wednesday = 60
   Thursday = 40
   Total children:$70 + 50 + 60 + 40 = 220$
   Boys = $\frac{1}{4}$of 220 = $\frac{1}{4} \times 220 = 55$
   Girls = Total – Boys= $220 - 55 = 165$

5. **Correct Answer: (B) 10**
   Solution: Each group of tally marks represents 5 objects. 25÷5=5.

6. **Correct Answer: (D) 270**
   P) Each 🚗 = 30 cars
   January = 8 × 30 = **240**
   March = 6 × 30 = **180**
   May = 5 × 30 = **150**
   $(240 + 180) - 150 = 270$
   Q) April = 4 × 30 = **120** cars
   Damaged = **25**
   Not damaged:$120 - 25 = 95$
   Fraction: $\frac{95}{120}$
   Answer: 95/120

7. **Correct Answer: B. 4**

   Given 6 calculator symbols = 24 calculators

   So, **1 calculator symbol** represents: $24 \div 6 = 4$

8. **Correct Answer: (A) 𝍸 ||**
   Solution: A group of 5 tallies is 𝍸. Adding 2 more makes 𝍸||.

9. **Correct Answer: (D) 20**
   Solution: Each 🎈 represents 2 balloons. 5×2=205 \times 2 = 205×2=20.

10. **Correct Answer: (A) Categories**
    Solution: The space between bars separates different categories.

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

11. **Correct Answer: (A) and (B)**
    Solution: Tally marks are grouped in 5s and are used for counting.

12. **Correct Answer: (A) and (D)**
    Solution: Bar graphs compare items and show trends clearly.

13. **Correct Answer: (A), (C), and (D)**
    Solution: Each 🏀 equals 3 balls, so 3×3=9.

14. **Correct Answer: (A), (B), and (D**
    Solution: A title, labeled axes, and a scale are essential for clarity.

15. **Correct Answer: (A), (B), and (C)**
    Solution: Based on the key, these calculations are correct.

ASSERTION AND REASON TYPE

16. **Correct Answer: (C)**

    Explanation: A pictograph uses pictures or symbols to represent data, but it is not restricted to data involving numbers above 1000. It can also represent smaller numbers.

17. **Correct Answer: (A)**

    Explanation: Grouping tally marks in sets of five helps simplify the counting process and reduces errors, making the explanation accurate.

18. **Correct Answer: (C)**

    Explanation: While bar graphs do compare categories, the bars in a bar graph are separated by spaces to distinguish categories. Therefore, the Reason is incorrect.

MATRIX MATCHING TYPE

19. **Correct Answer: (a) → (Q), (b) → (R), (c) → (P), (d) → (S)**
Explanation:
    - Each symbol in a pictograph corresponds to a specific quantity, so it helps calculate the total items.
    - A key explains what each symbol represents, aiding understanding.
    - A symbol representing a fixed value makes it easier to scale the data.
    - Pictographs visually represent data, making it easier to understand.
20. **Correct Answer: (a) → (P), (b) → (Q), (c) → (R), (d) → (S)**
Explanation:
    - Four vertical lines with a slash form a group of five tally marks.
    - Tally marks are an easy method to count items or events.
    - Tally marks are commonly used for recording repetitive data like attendance.
    - Grouping in fives ensures fewer counting errors.

COMPREHENSION TYPE

Passage-1

21. **Correct Answer: B) ₹50**
(5 apples × ₹10 = ₹50)
22. **Correct Answer: C) 6**
The pictograph shows **6 bananas**.
23. **Correct Answer: A) Apples**
Apples = $\mathbf{5 \times 10 = 50}$
    - Bananas = $\mathbf{6 \times 5 = 30}$
    - Oranges = $\mathbf{7 \times 7 = 49}$

    Highest = ₹50

Passage-2

24. **Correct Answer: C) Football**
The highest number is **10**, which is for **Football**.
25. **Correct Answer: B) 12**
Swimming = **6**
Basketball = **6**
$$6 + 6 = 12$$
26. **Correct Answer: B) 2**
Football = **10**
Cricket = **8**
$$10 - 8 = 2$$

8. Shapes and Patterns

Solutions

Worksheet – 1

PICTOGRAPH

1. **Correct Answer: b) Triangle**
   Solution: A triangle is a 2D shape with three sides and three corners.
2. **Correct Answer: b) Square**
   Solution: A square has four equal sides and four right angles.
3. **Correct Answer: b) Cube**
   Solution: A cube is a 3D shape with six equal square faces.
4. **Correct Answer: b) Green**
   Solution: Mixing blue and yellow creates green.
5. **Correct Answer: C) Red and sphere**
   Solution: A ball is round (sphere) and red in colour.
6. **Correct Answer: C) Yellow**
   Solution: Yellow is one of the primary colours that cannot be made by mixing other colours.
7. **Correct Answer: C) Circle**
   Solution: A circle has no sides or corners and is a smooth, round shape.

Worksheet-2

1. **Correct Answer: a)** 
   Solution: The unit of repeat is the smallest part of the pattern, which is .
2. **Correct Answer: b)** 
   Solution: The pattern alternates between  and  , so the next part would be  .
3. **Correct Answer: b)** 
   Solution: This pattern grows because it keeps adding more of the same color, so the number of  increases.
4. **Correct Answer: a)** 
   Solution: The unit of repeat is , as the pattern repeats this sequence multiple times.
5. **Correct Answer: c)** 
   Solution: The pattern shrinks because it starts with four red circles and decreases by one circle each time.
6. **Correct Answer: c) It increases**
   Solution: In a growing pattern, the number of objects increases at each step.
7. **Correct Answer: a)** 
   Solution: The pattern alternates between  and  , so the next object is  .

Worksheet-3

1. **Correct Answer: a) 5**
   Solution: The pattern is increasing by 1 each time, so the next number is 5.
2. **Correct Answer: b) 2, 4, 6, 8, 10**
   Solution: The pattern is increasing by 2 each time, which is skip counting by 2.
3. **Correct Answer: a) 25**
   Solution: The pattern increases by 5 each time, so the next number is 25.
4. **Correct Answer: c) 2, 4, 8, 16**
   Solution: This pattern follows multiplication by 2, where each number is multiplied by 2 to get the next one.
5. **Correct Answer: b) 15**
   Solution: The pattern increases by 1, 2, 3, 4, and the next number will increase by 5, making it 15
6. **Correct Answer: c) 33**
   Solution: The pattern is increasing by 3 each time, so the next number is 33.
7. **Correct Answer: b) Addition by 5**
   Solution: The pattern increases by 5 each time.

Worksheet-4

1. **Correct Answer: b) A pattern made of shapes that fit together perfectly**

   Solution: A tessellation is a repeating pattern where the shapes fit together with no gaps.

2. **Correct Answer: b) A tiled floor**

   Solution: The floor tiles are arranged in a way that they fit together perfectly without gaps, which is a tessellation.

3. **Correct Answer: c) Any shape that fits together without gaps**

   Solution: Tessellations can be made with many shapes, as long as they fit together to cover a space without leaving any gaps.

4. **Correct Answer: b) A pattern of triangles and squares**

   Solution: Regular tessellations use only one type of shape. A pattern with triangles and squares is a semi-regular tessellation, not a regular one.

5. **Correct Answer: b) Hexagon**

   Solution: The honeycomb cells are hexagonal and fit together perfectly in a tessellation.

6. **Correct Answer: b) A beehive**

   Solution: A beehive is a great example of tessellation, as the bees create hexagonal shapes that fit together perfectly.

7. **Correct Answer: b) The shapes fit together without gaps**

   Solution: The key feature of a tessellation is that the shapes fit together perfectly with no gaps between them.

Worksheet-5

1. **Correct Answer: b) When two halves of a shape are the same**

   Solution: Symmetry is when one half of an object matches the other half exactly, like folding paper and seeing both sides match.

2. **Correct Answer: b) A butterfly**

   Solution: A butterfly has line symmetry because its wings are the same on both sides when folded along the center.

3. **Correct Answer: b) The two halves match perfectly**

   Solution: A heart shape has line symmetry, so if you fold it in half, the two halves will look the same.

4. **Correct Answer: c) A star**

   Solution: A star has rotational symmetry because it looks the same after being rotated at certain angles.

5. **Correct Answer: c) A circle**

   Solution: A circle has point symmetry because every point on the circle is equidistant from the center point.

6. **Correct Answer: d) A flower with equally spaced petals**

   Solution: Flowers like daisies have line symmetry, where each petal mirrors the other.

7. **Correct Answer: b) When you can rotate a shape and it looks the same**

   Solution: Rotational symmetry happens when rotating a shape around a point makes it look the same at certain angles.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: (C) Square**

   Explanation: A square has four equal sides and four right angles.

2. **Correct Answer: (B) 2**

   Explanation: A rectangle has two lines of symmetry: one vertical and one horizontal.

3. **Correct Answer: (B)** 

   Explanation: The pattern repeats every two shapes:  .

4. **Correct Answer: B) 15**

   Solution: The pattern increases by 3 each time.
   3 → 6 → 9 → 12 → 15

5. **Correct Answer: A)** ★★★★

   Solution: The stars increase by **one in each row**:
   1 star → 2 stars → 3 stars → **4 stars**
   So the next should be **4 stars**.

6. **Correct Answer: (C) 6 lines**

   Explanation: A regular hexagon has six lines of symmetry.

7. **Correct Answer: (B) Two equal horizontal halves**

   Explanation: A horizontal line of symmetry divides the shape into two equal horizontal halves.

8. **Correct Answer: D) Triangle**

   Solution: Cube, cylinder, and sphere are **3D shapes**, but triangle is a **2D shape**.

9. **Correct Answer:** C) Rectangle

   Explanation: A break it looks like an rectangle then the tessellation is rectangle

10. **Correct Answer: (C) Infinite**

    Explanation: A circle has infinite lines of symmetry, as any line passing through its center divides it into two equal halves.

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

11. **Correct Answer:**

    (A) Square and **(B) Circle**

    Explanation: Squares and circles have lines of symmetry; stars and triangles depend on their type.

12. **Correct Answer: (B) They can repeat and (C) They can shrink**

    Explanation: Patterns can grow, shrink, or repeat and may not involve numbers.

13. **Correct Answer: (A)** ▲**, (B)** ◆**, and (D)** ▭

    Explanation: Triangles, hexagons, and rectangles tessellate, but circles do not.

14. **Correct Answer: (A)** ●**,** ●●**,** ●●● **and (B) 2, 4, 6, 8**

    Explanation: These patterns grow in quantity or size.

15. **Correct Answer: (B) Square, (C) Rectangle, and (D) Hexagon**

    Explanation: A square has 4 lines, a rectangle 2 lines, and a hexagon 6 lines of symmetry.

ASSERTION AND REASON TYPE

16. **Correct Answer: (B)**

    Explanation: A square is a type of rectangle because it has opposite sides equal and all angles are right angles. However, a square's defining feature is that all sides are equal, which makes the Reason true but not an explanation of the Assertion.

17. **Correct Answer: (A)**

    Explanation: Patterns rely on a unit of repeat to determine the sequence. The unit of repeat forms the base of the pattern and is essential for understanding the rule.

18. **Correct Answer: (D)**

    Explanation: Circles cannot tessellate because their curved edges leave gaps between them. While the Reason is true (circles are round), it does not support the incorrect Assertion.

MATRIX MATCHING TYPE

19. **Correct Answer:**

    Answer Key:

    (a) → Circle ●

    (b) → Alternating Pattern ⚽ ● ⚽ ●

    (c) → Pentagon ▲

    (d) → Growing Patterns

    Explanation:

    1. A **circle** has no corners.
    2. An **alternating pattern** repeats two items in order, like red and blue balls.
    3. A **pentagon** is a five-sided shape.

20. Correct Answer:

Growing patterns increase in size or quantity, like plants growing

Answer Key:

- → Square
- → Tessellation
- → Number Pattern
- → Symmetry

Explanation:

1. A **square** has four equal sides.
2. **Tessellation** is a way of arranging shapes without gaps, like tiles on a floor.
3. A **number pattern** increases or decreases by a constant number.
4. **Symmetry** means an object can be divided into two identical parts.

COMPREHENSION TYPE

21. Correct Answer: B) 5

Solution: The number of flowers increases by 1 in each row.
1st = 1, 2nd = 2, 3rd = 3, 4th = 4, 5th = 5

22. Correct Answer: C) 55

Solution: Add flowers from row 1 to row 10
1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 = 55

23. Correct Answer: B) Growing pattern

Solution: The number is increasing row by row, so it is a growing pattern.

Passage-2

24. Correct Answer: C) Hexagon

What is the shape used in the tessellation?
Solution: The passage clearly says each tile was a hexagon.

25. Correct Answer: B) No, they leave gaps

Can circles tessellate like hexagons? Why or why not?
Solution: Circles do not fit together perfectly and leave gaps.

26. Correct Answer: C) no gaps or overlaps

Tessellation means shapes fit together with
Solution: In tessellation, shapes fit perfectly with no gaps or overlaps.